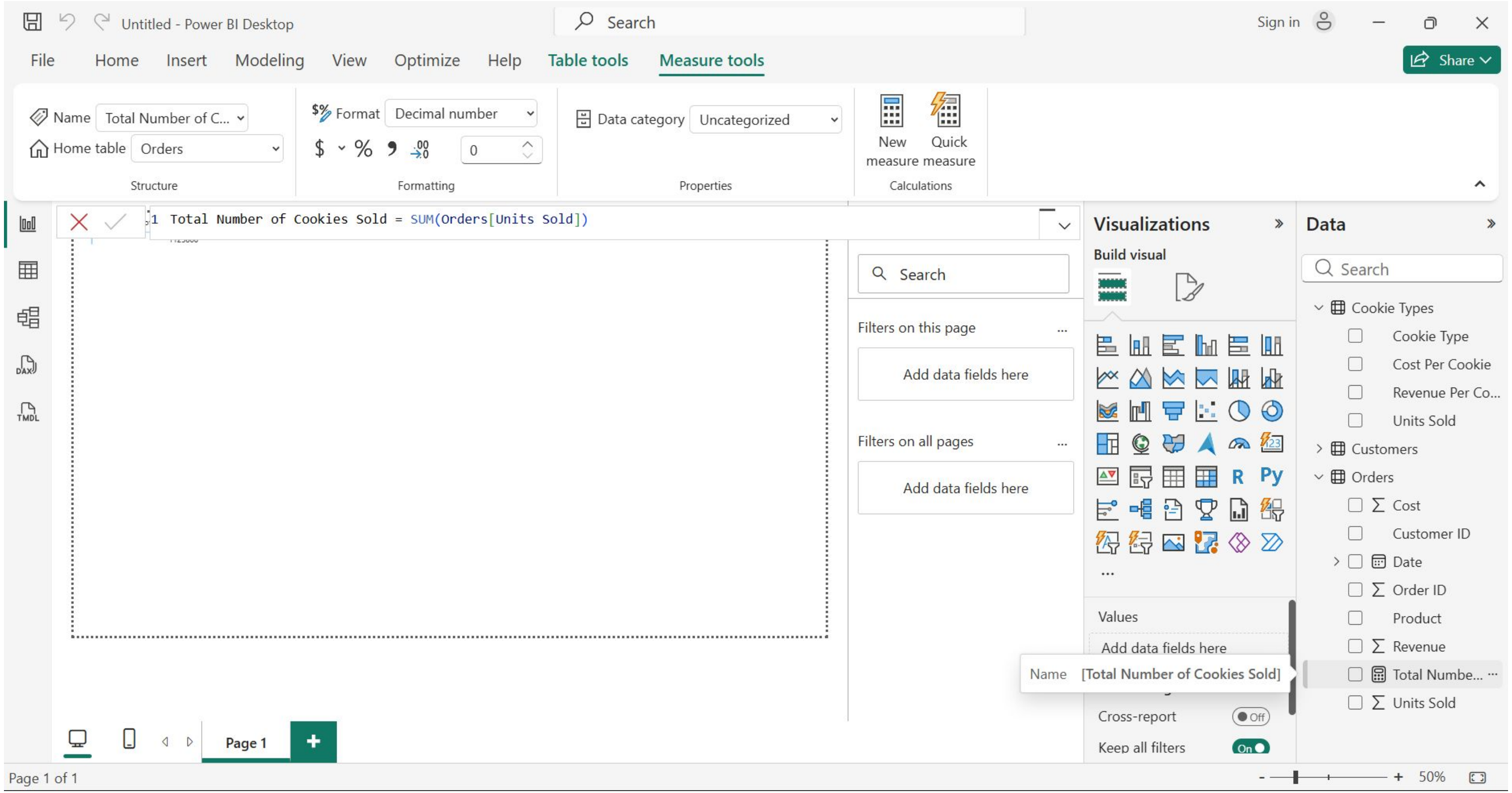
Applying DAX
Fundamentals in Power BI: From Measures to Contextual Analysis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1 Total Number of Cookies Sold = SUM(Orders[Units Sold])
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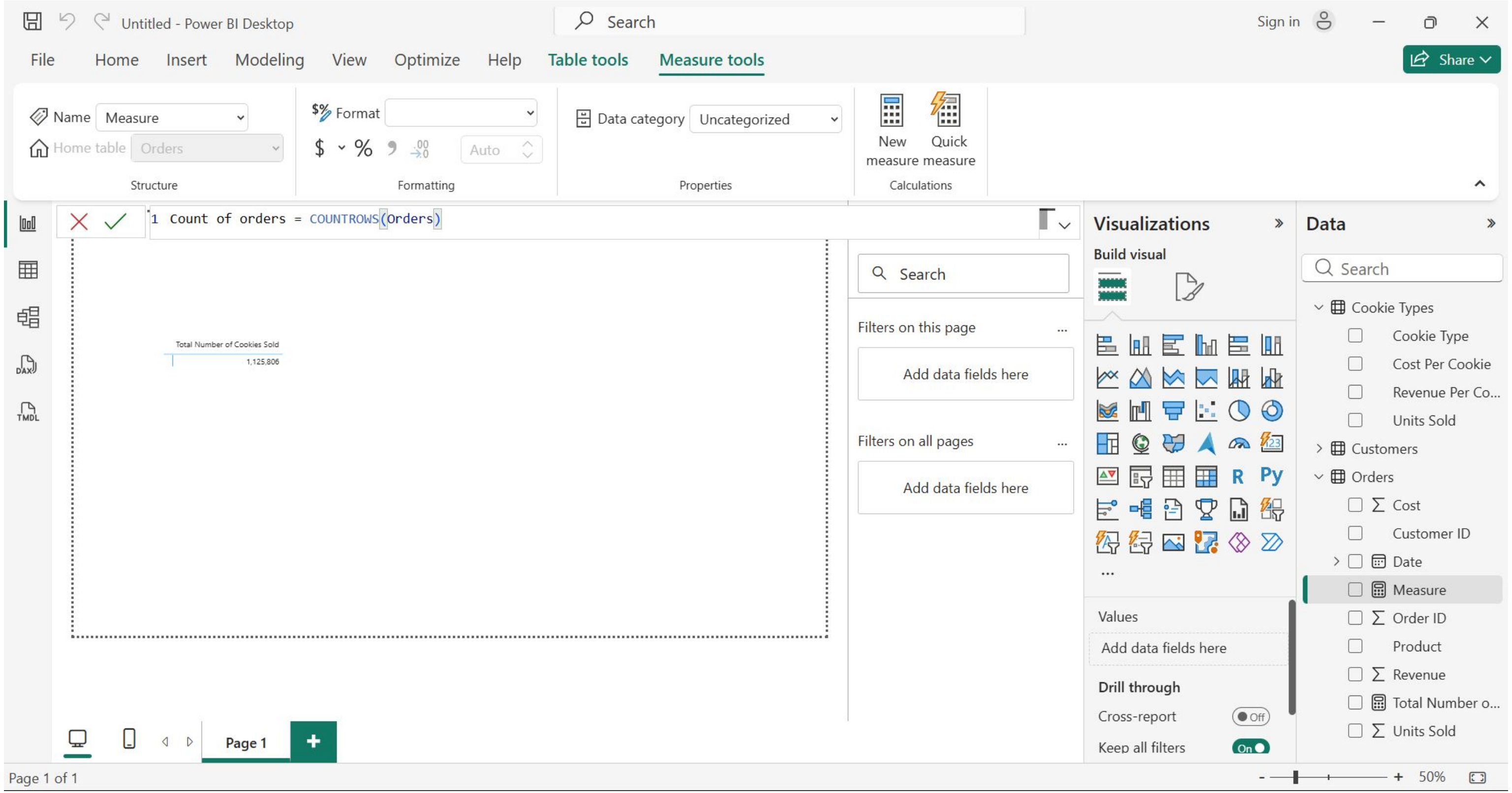
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Calculations
1 Count of orders = COUNTROWS(Orders)
Total Number of Cookies Sold
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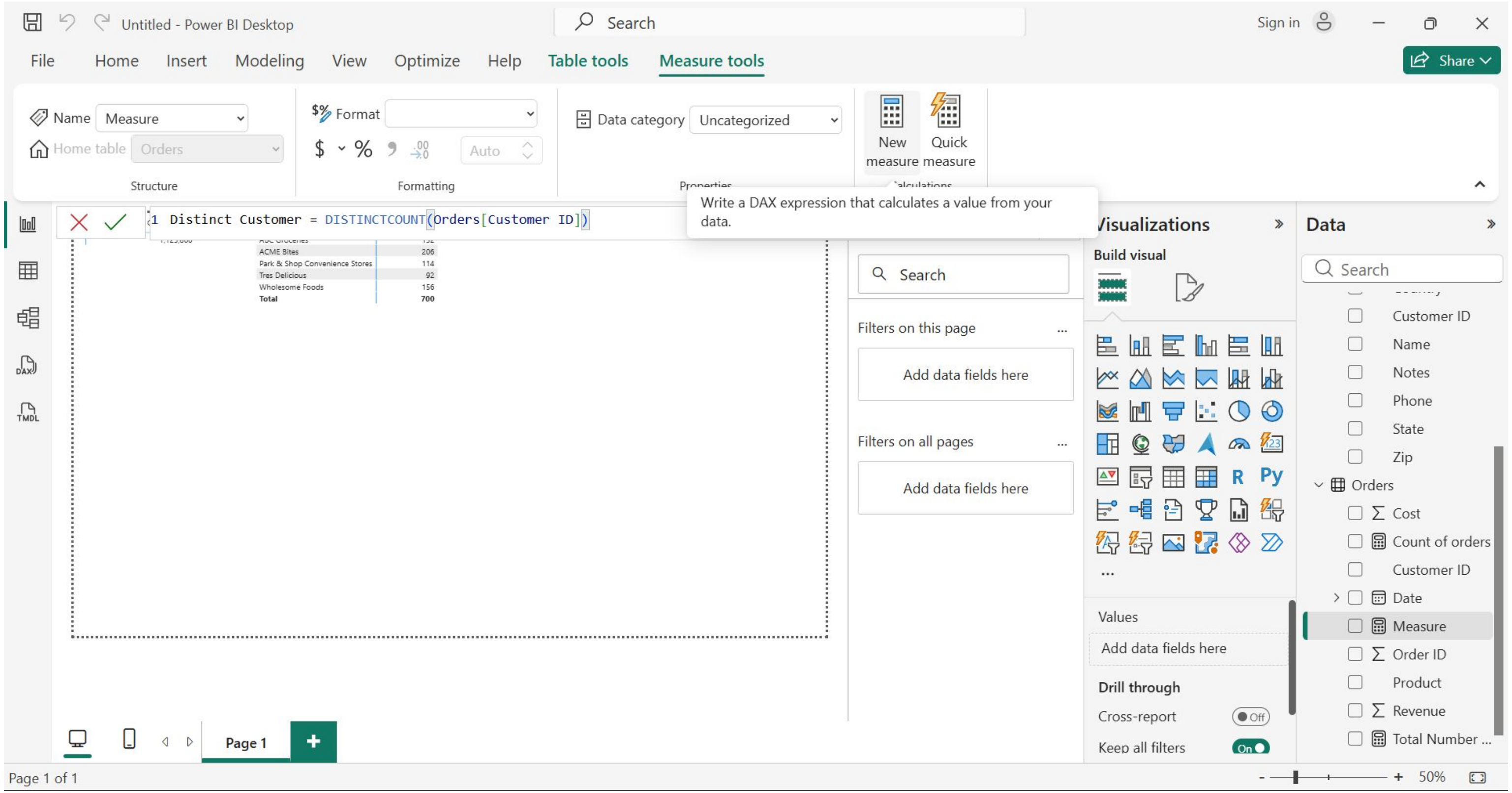
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Home table
Orders
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Auto
Formatting
Data category
Uncategorized
Properties
New measure
Quick measure
Calculations
1 Distinct Customer = DISTINCTCOUNT(Orders[Customer ID])
Write a DAX expression that calculates a value from your data.
ACME Bites
206
Park & Shop Convenience Stores
114
Tres Delicious
92
Wholesome Foods
156
Total
700
Search
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
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Values
Add data fields here
Drill through
Cross-report
Off
Keep all filters
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Data
Search
Customer ID
Name
Notes
Phone
State
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Orders
Cost
Count of orders
Customer ID
Date
Measure
Order ID
Product
Revenue
Total Number ...
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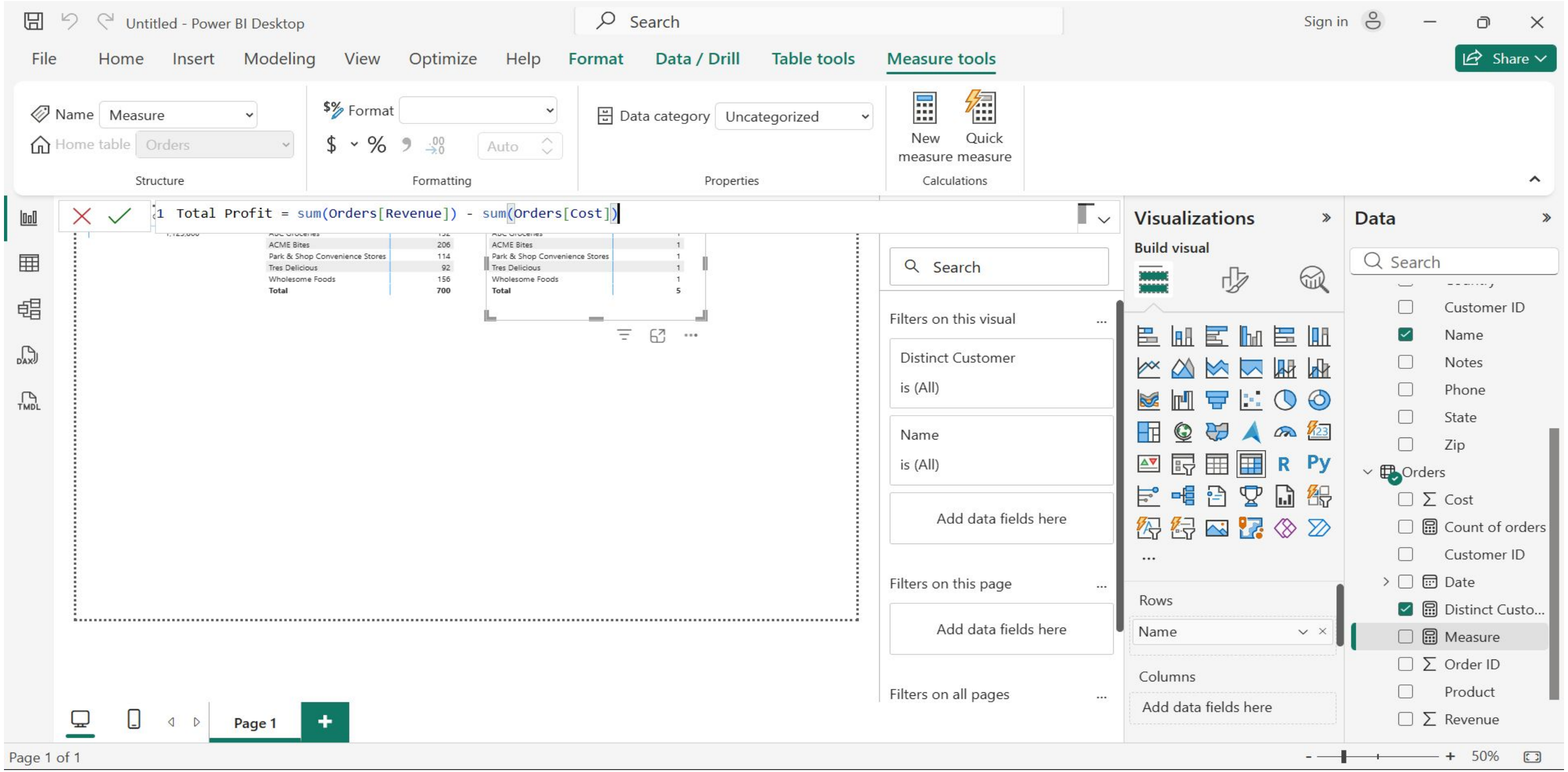
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Properties
New measure
Quick measure
Calculations
1 Total Profit = sum(Orders[Revenue]) - sum(Orders[Cost])
ACME Bites
Park & Shop Convenience Stores
Tres Delicious
Wholesome Foods
Total
Visualizations
Build visual
Filters on this visual
Distinct Customer
is (All)
Name
is (All)
Add data fields here
Filters on this page
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Filters on all pages
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Name
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Add data fields here
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State
Zip
Orders
Cost
Count of orders
Customer ID
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Measure
Order ID
Product
Revenue
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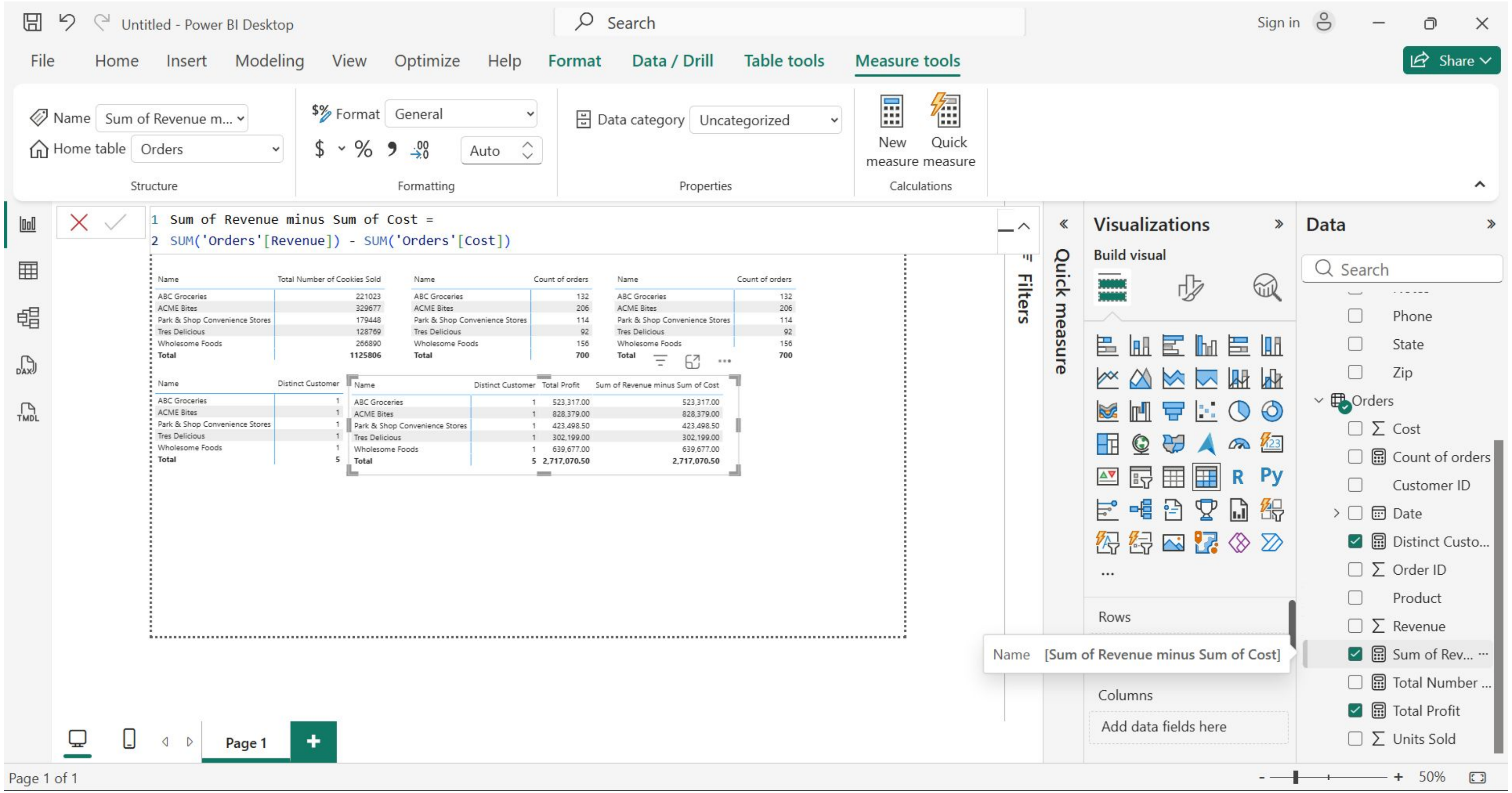
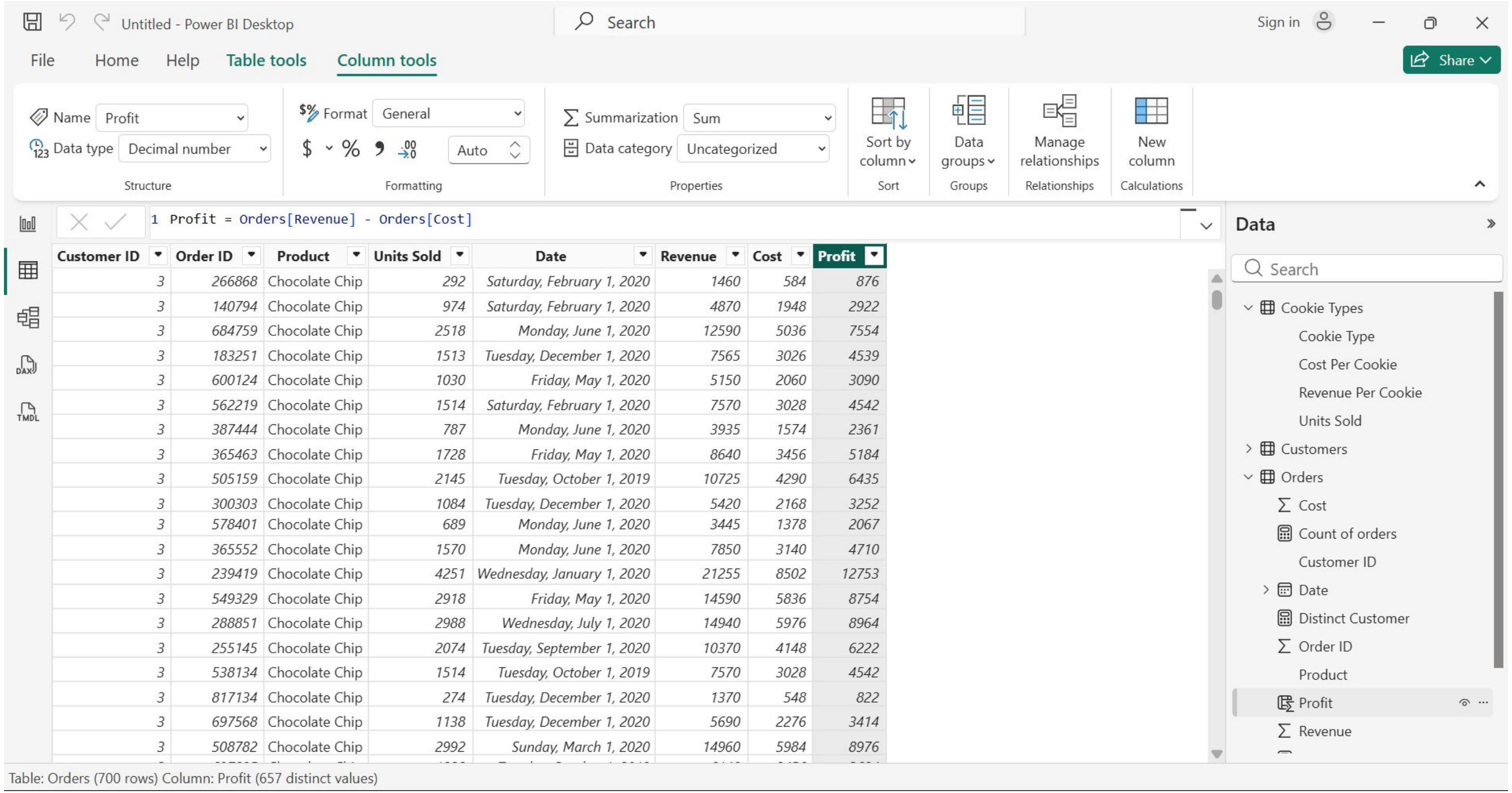
Using Quick Measure:

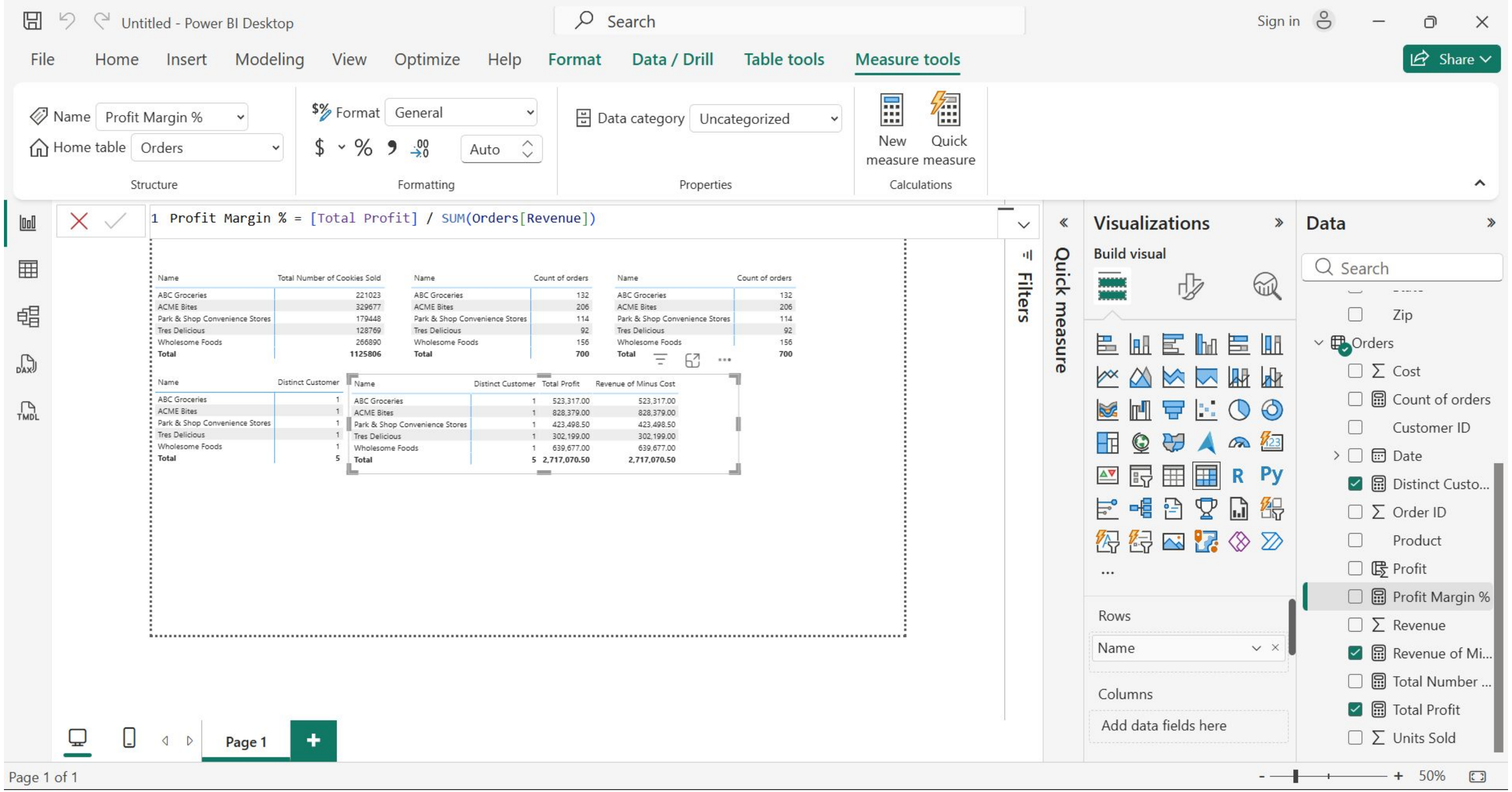


Profit:

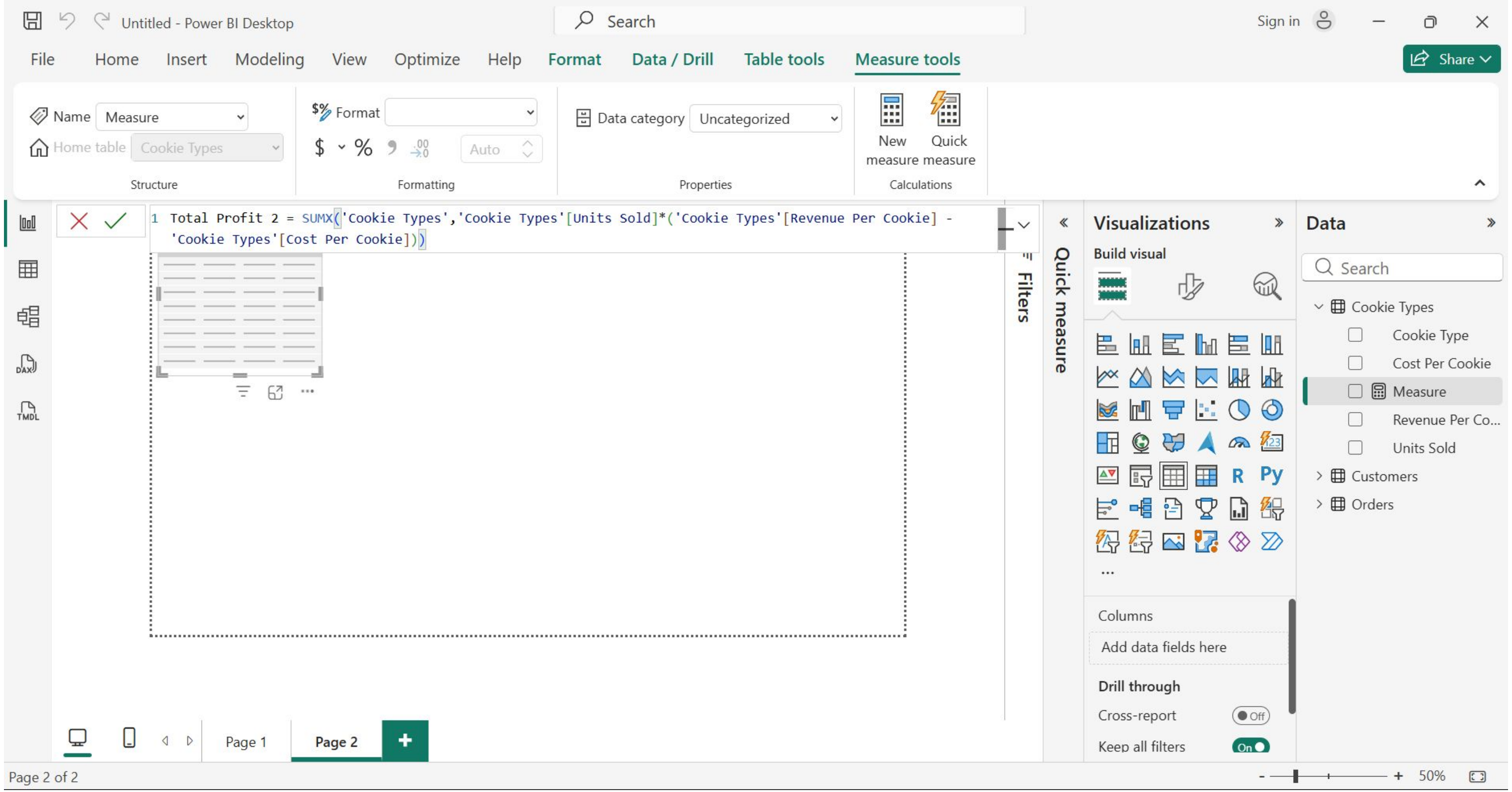


Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036	7554
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026	4539
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060	3090
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028	4542
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574	2361
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456	5184
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290	6435
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168	3252
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378	2067
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140	4710
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502	12753
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836	8754
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976	8964
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148	6222
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028	4542
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548	822
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276	3414
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984	8976

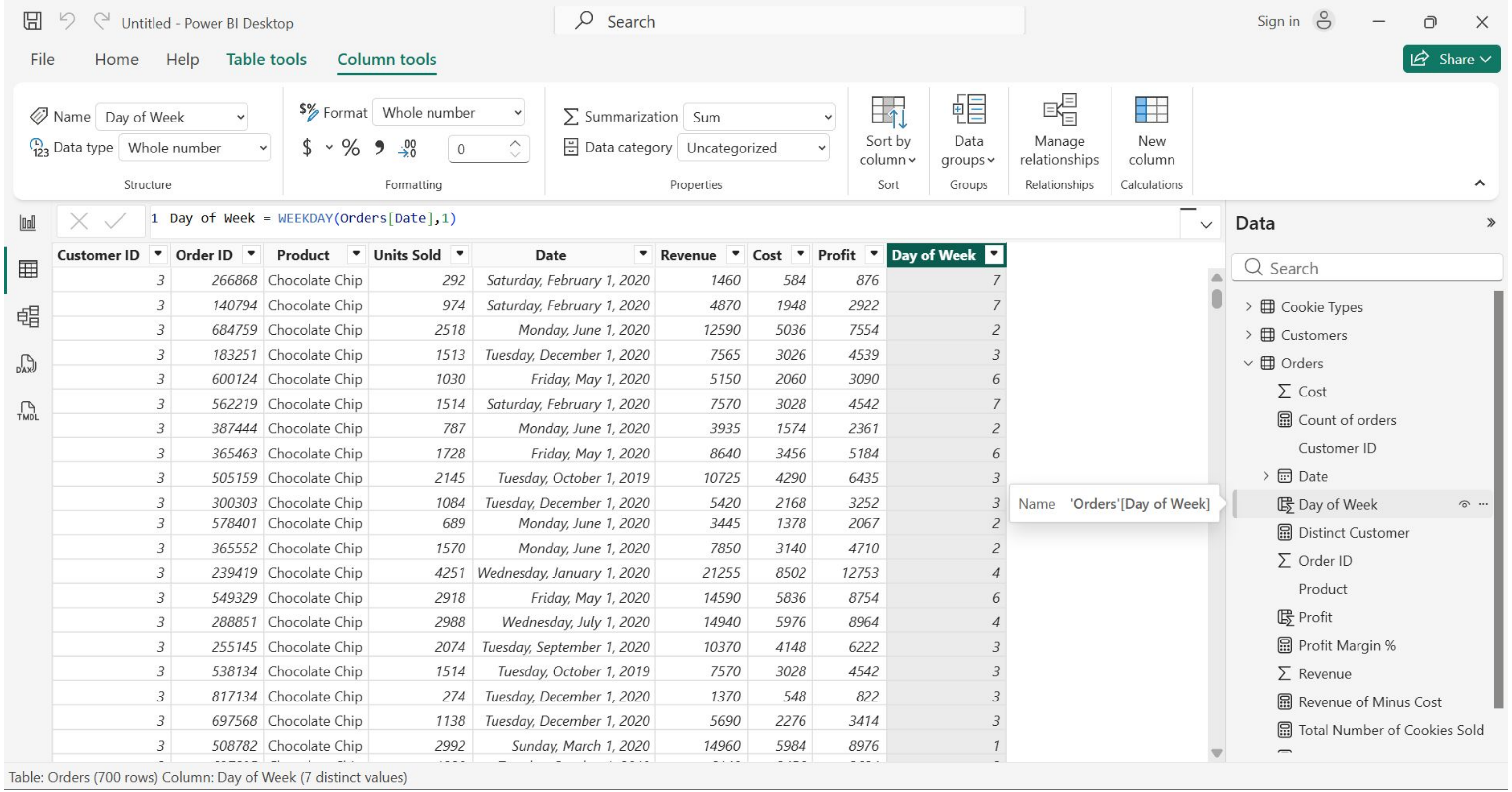
Profit Margin %:



Cookie Types:



Time and date Function:

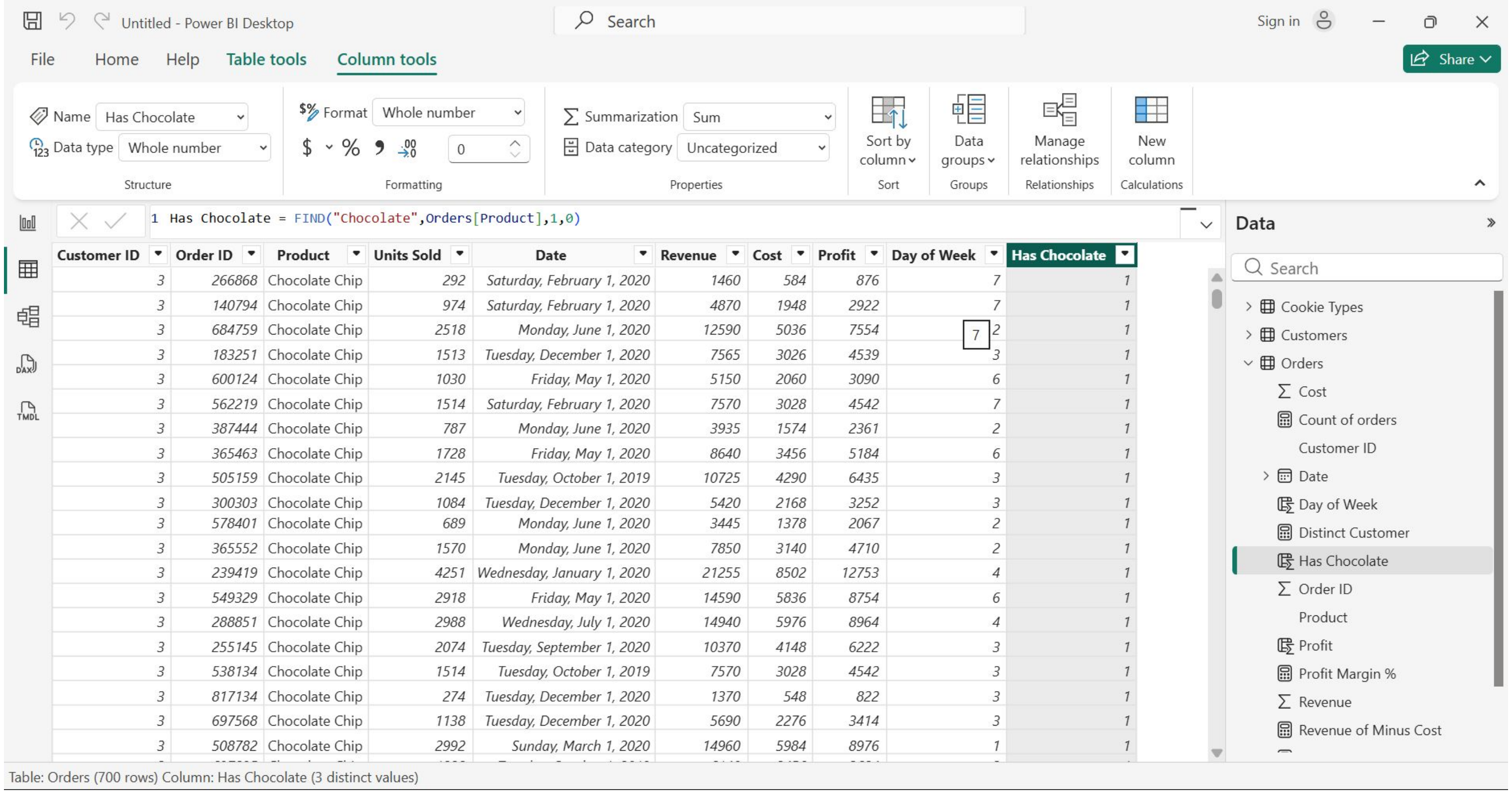



```
1 Day of Week = WEEKDAY(Orders[Date],1)
```

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit	Day of Week
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876	7
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922	7
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036	7554	2
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026	4539	3
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060	3090	6
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028	4542	7
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574	2361	2
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456	5184	6
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290	6435	3
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168	3252	3
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378	2067	2
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140	4710	2
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502	12753	4
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836	8754	6
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976	8964	4
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148	6222	3
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028	4542	3
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548	822	3
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276	3414	3
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984	8976	1

Table: Orders (700 rows) Column: Day of Week (7 distinct values)

Find:



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Name Has Chocolate
Data type Whole number
Structure
Format Whole number
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Data category Uncategorized
Properties
Sort by column
Sort
Data groups
Groups
Manage relationships
Relationships
New column
Calculations
1 Has Chocolate = FIND("Chocolate",Orders[Product],1,0)
Customer ID Order ID Product Units Sold Date Revenue Cost Profit Day of Week Has Chocolate
3 266868 Chocolate Chip 292 Saturday, February 1, 2020 1460 584 876 7 1
3 140794 Chocolate Chip 974 Saturday, February 1, 2020 4870 1948 2922 7 1
3 684759 Chocolate Chip 2518 Monday, June 1, 2020 12590 5036 7554 2 1
7
3 183251 Chocolate Chip 1513 Tuesday, December 1, 2020 7565 3026 4539 3 1
3 600124 Chocolate Chip 1030 Friday, May 1, 2020 5150 2060 3090 6 1
3 562219 Chocolate Chip 1514 Saturday, February 1, 2020 7570 3028 4542 7 1
3 387444 Chocolate Chip 787 Monday, June 1, 2020 3935 1574 2361 2 1
3 365463 Chocolate Chip 1728 Friday, May 1, 2020 8640 3456 5184 6 1
3 505159 Chocolate Chip 2145 Tuesday, October 1, 2019 10725 4290 6435 3 1
3 300303 Chocolate Chip 1084 Tuesday, December 1, 2020 5420 2168 3252 3 1
3 578401 Chocolate Chip 689 Monday, June 1, 2020 3445 1378 2067 2 1
3 365552 Chocolate Chip 1570 Monday, June 1, 2020 7850 3140 4710 2 1
3 239419 Chocolate Chip 4251 Wednesday, January 1, 2020 21255 8502 12753 4 1
3 549329 Chocolate Chip 2918 Friday, May 1, 2020 14590 5836 8754 6 1
3 288851 Chocolate Chip 2988 Wednesday, July 1, 2020 14940 5976 8964 4 1
3 255145 Chocolate Chip 2074 Tuesday, September 1, 2020 10370 4148 6222 3 1
3 538134 Chocolate Chip 1514 Tuesday, October 1, 2019 7570 3028 4542 3 1
3 817134 Chocolate Chip 274 Tuesday, December 1, 2020 1370 548 822 3 1
3 697568 Chocolate Chip 1138 Tuesday, December 1, 2020 5690 2276 3414 3 1
3 508782 Chocolate Chip 2992 Sunday, March 1, 2020 14960 5984 8976 1 1
Data
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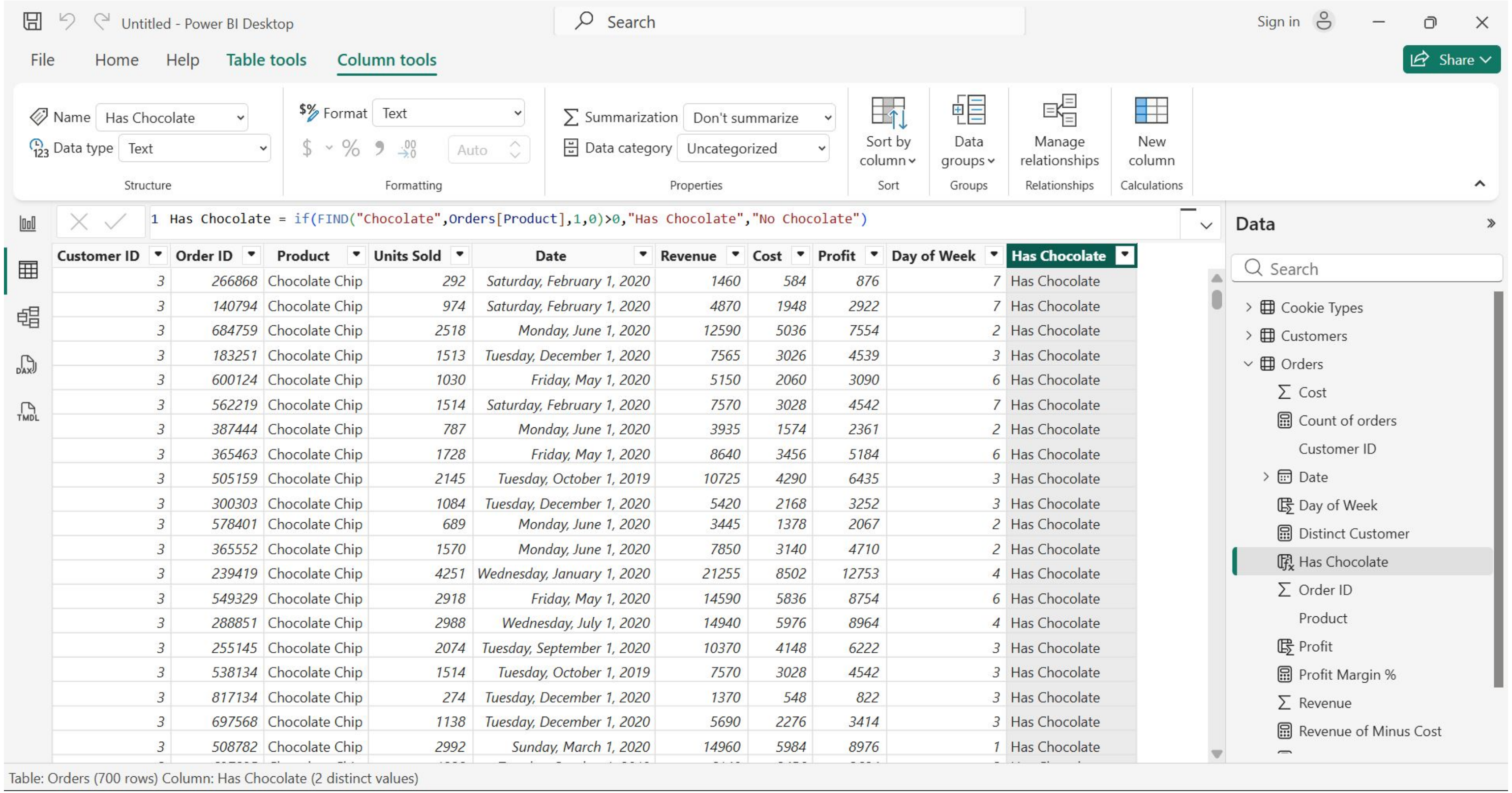
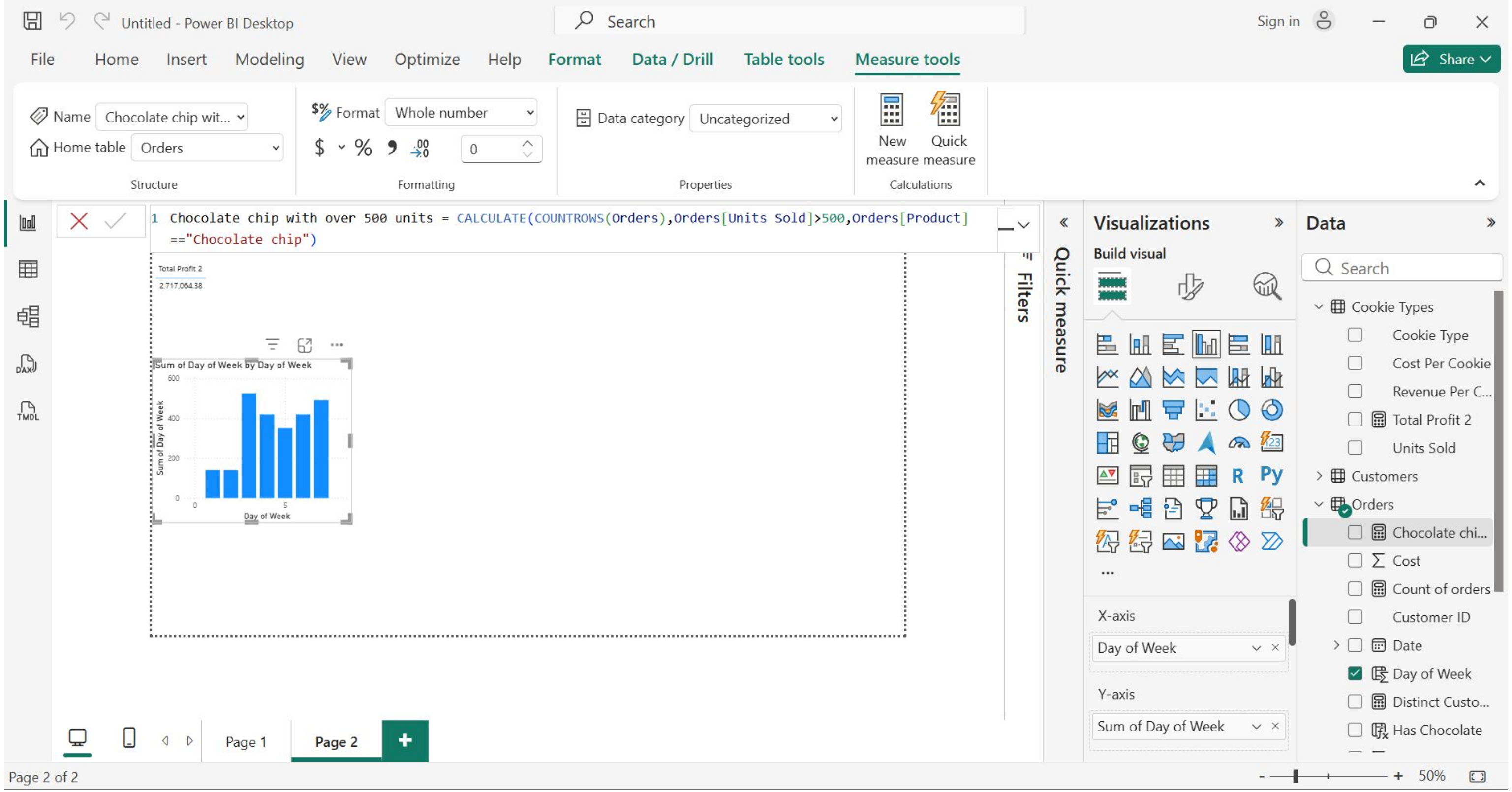
Cookie Types
Customers
Orders
Cost
Count of orders
Customer ID
Date
Day of Week
Distinct Customer
Has Chocolate
Order ID
Product
Profit
Profit Margin %
Revenue
Revenue of Minus Cost
Table: Orders (700 rows) Column: Has Chocolate (3 distinct values)

If Statement:

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New measure
Quick measure
Calculations
1 Chocolate chip with over 500 units = CALCULATE(COUNTROWS(Orders),Orders[Units Sold]>500,Orders[Product]
=="Chocolate chip")
Total Profit 2
2,717,064.38
Sum of Day of Week by Day of Week
Sum of Day of Week
Day of Week
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Has Chocolate
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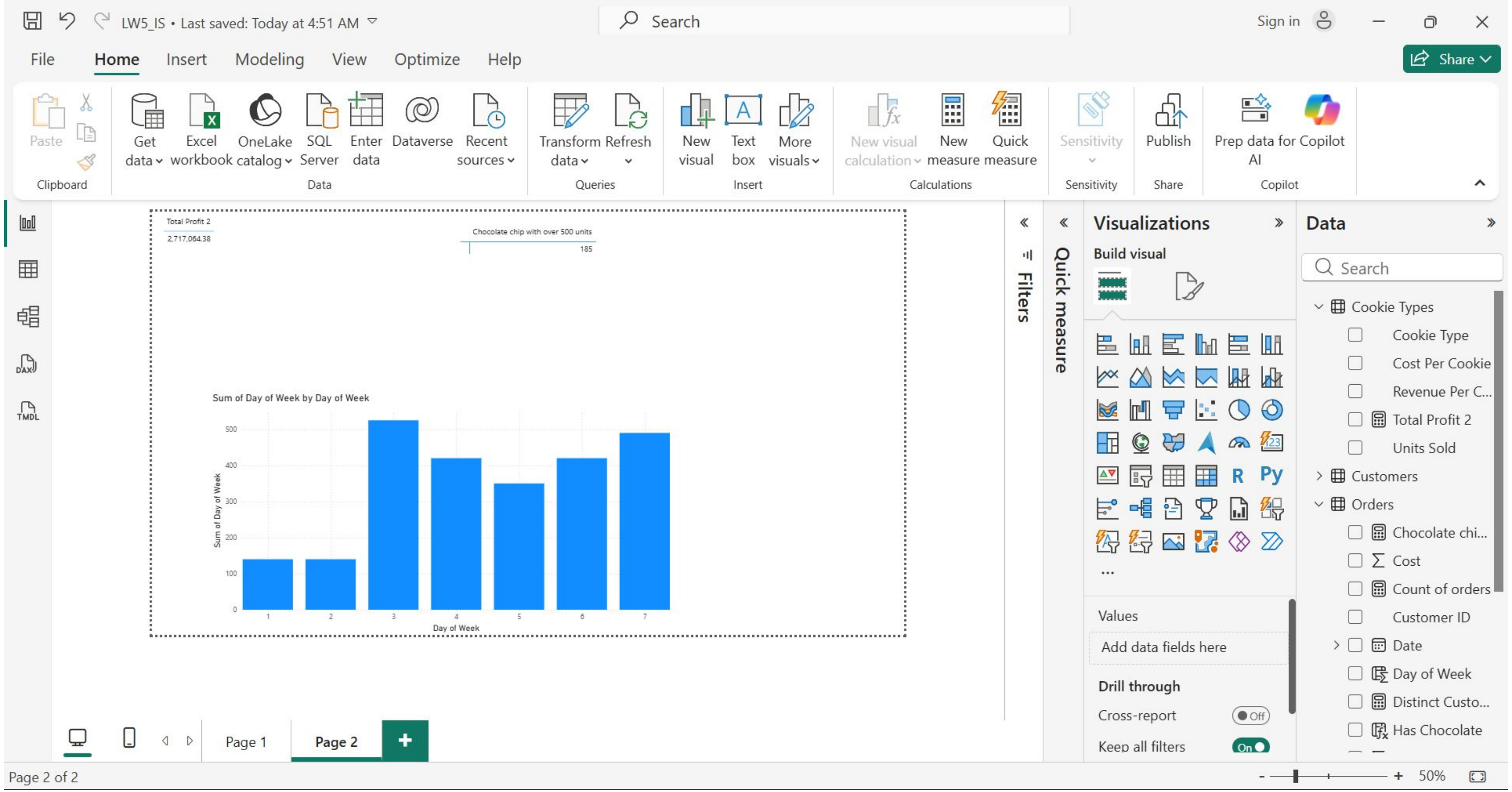

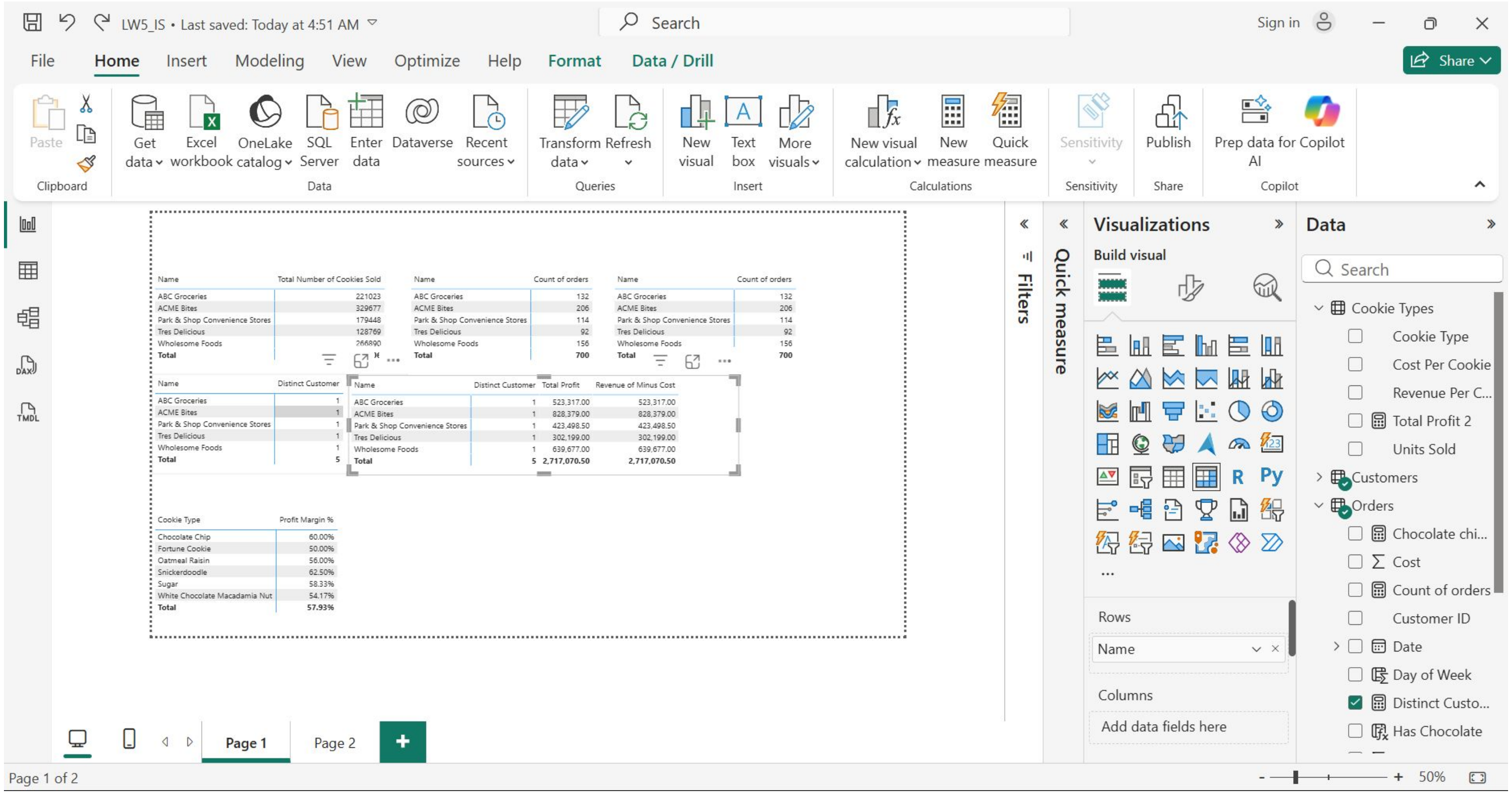

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Name
Total Number of Cookies Sold
ABC Groceries 221023
ACME Bites 329677
Park & Shop Convenience Stores 179448
Tres Delicious 128769
Wholesome Foods 266890
Total
Name
Count of orders
ABC Groceries 132
ACME Bites 206
Park & Shop Convenience Stores 114
Tres Delicious 92
Wholesome Foods 156
Total 700
Name
Count of orders
ABC Groceries 132
ACME Bites 206
Park & Shop Convenience Stores 114
Tres Delicious 92
Wholesome Foods 156
Total 700
Name
Distinct Customer
ABC Groceries 1
ACME Bites 1
Park & Shop Convenience Stores 1
Tres Delicious 1
Wholesome Foods 1
Total 5
Name
Distinct Customer
Total Profit
Revenue of Minus Cost
ABC Groceries 1 523,317.00 523,317.00
ACME Bites 1 828,379.00 828,379.00
Park & Shop Convenience Stores 1 423,498.50 423,498.50
Tres Delicious 1 302,199.00 302,199.00
Wholesome Foods 1 639,677.00 639,677.00
Total 5 2,717,070.50 2,717,070.50
Cookie Type
Profit Margin %
Chocolate Chip 60.00%
Fortune Cookie 50.00%
Oatmeal Raisin 56.00%
Snickerdoodle 62.50%
Sugar 58.33%
White Chocolate Macadamia Nut 54.17%
Total 57.93%
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Cookie Type
Cost Per Cookie
Revenue Per C...
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Units Sold
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Has Chocolate
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Guided Questions:

1. Defining DAX: DAX, or Data Analysis Expressions, works differently. Instead of thinking about single cells, DAX works with entire tables and relationships between data. It helps Power BI analyze large amounts of information and create calculations that automatically adjust based on filters, charts, or user selections in a report.

2. The “Why” of DAX

   Raw data alone is usually not enough to create meaningful reports. Imported datasets often only contain basic information such as sales amounts, dates, products, or customer names. They do not automatically provide deeper business insights.

GUIDE QUESTIONS:

3. The Storage Difference

A Calculated Column increases the file size of a Power BI model because the calculated values are stored inside the dataset itself. Power BI computes the result for every row and saves those results in memory.

4. Context Clues – How Filter Context Affects a Measure

Filter Context acts like a “lens” that tells the Measure which portion of the data it should calculate from at that moment.

That is why Measures are powerful in dashboards—they create interactive reports where numbers automatically adjust as users explore the data.

5. The Related Function:

The RELATED function is necessary when working with multiple tables because the data in Power BI is separated into different tables such as Orders, Customers, and Cookie Types. The function allows a column from one table to be accessed and used in another related table.

For example, if the Orders table contains a CustomerID, the RELATED function can retrieve the customer name from the Customers table so it can be used in calculations or reports.

For the RELATED function to work correctly, a relationship between the tables must first be established in the Model View. Usually, this relationship is based on matching fields like CustomerID or CookieTypeID. Without these relationships, Power BI cannot determine how the tables are connected.

6. Deconstructing the Formula

The formula creates a measure named Total Sales that calculates the total of all values found in the Sales Amount column from the Sales table using the SUM function.

Including the table name in the reference (Sales[Sales Amount]) is considered best practice because it makes the formula clearer and easier to understand. It also avoids confusion when different tables contain columns with similar names and helps organize calculations properly in models with multiple related tables.

7. Data Interpretation

If the Total Profit measure shows a negative number for a specific category, it does not automatically mean that the DAX formula is broken. In most cases, it means the business data shows a loss for that category, where costs are higher than revenue or sales.

If the calculations match the raw data, then the DAX formula is working correctly and the negative value reflects actual business performance.

8. Optimization

It would be more efficient to create a single Measure for Profit Margin Percentage instead of a Calculated Column for every row.

Calculated columns store a value for every row in the table, which increases file size and can slow down the model. Since Profit Margin Percentage is usually used for analysis and reporting, a measure is the better choice in DAX.